

Educational Codeforces Round 189 (Rated for Div. 2)

A. A Number Between Two Others

2 seconds, 512 megabytes

You are given two integers x and y such that $y > x$ and $y \bmod x = 0$ (that is, y is divisible by x).

Your task is to determine whether there exists an integer z such that

- z lies between x and y (that is, $x < z < y$);
- z is divisible by x (that is, $z \bmod x = 0$);
- y is not divisible by z (that is, $y \bmod z \neq 0$).

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

Each test case consists of a single line containing two integers x and y ($1 \leq x < y \leq 10^{18}$; $y \bmod x = 0$).

Output

For each test case, print the answer as follows: if the required number z exists, print YES; otherwise, print NO. You may print each letter in any case.

input
5
1 2
1 3
1234567890 12345678900
2 8
7 84
output
NO
YES
YES
YES
YES

In the second test case of the example, you can use $z = 2$.

In the third test case of the example, you can use $z = 7407407340$.

B. Alternating String

2 seconds, 512 megabytes

Let us call a string t *alternating* if for every i from 1 to $n - 1$, the condition $t_i \neq t_{i+1}$ holds.

You are given a string s consisting only of the letters "a" and/or "b". You may perform the following operation on it **at most once**:

- choose a substring $s_l s_{l+1} \dots s_r$ consisting of at least one character;
- after choosing the substring, you may invert all letters in it, that is, change every "a" to "b" and every "b" to "a" (you are allowed to do it, but you don't have to do it);
- reverse the chosen substring, that is, transform the string $s_1 s_2 \dots s_{l-1} s_l s_{l+1} \dots s_r s_{r+1} \dots s_n$ into $s_1 s_2 \dots s_{l-1} s_r s_{r-1} \dots s_l s_{l+1} \dots s_n$.

Note that when performing such an operation, you are not required to do the second step. For example, for the string $s = \text{"ababbab"}$, after one operation you can obtain "abababa" by choosing the substring $s_5 s_6 s_7$ and doing the second step, or "bababab" by choosing the substring $s_1 s_2 s_3 s_4$ and not doing the second step. However, you always have to perform the third step of the operation.

Your task is to determine whether it is possible to obtain any *alternating* string from the string s .

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

Each test case consists of one line containing one string s ($2 \leq |s| \leq 2 \cdot 10^5$). The string consists only of lowercase Latin letters "a" and/or "b".

Additional constraints on the input:

- the sum of string lengths over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output one of the following:

- YES, if it is possible to obtain an *alternating* string from s ;
- NO otherwise.

You may output each letter in any case.

input
8
abbaba
aaaaa
bababba
ab
abbaabba
bbb
ababa
aabb
output
YES
NO
YES
YES
NO
YES
YES
YES

In the first example, you can choose the substring $s_3 s_4 s_5 s_6$ and skip the second step; then you get the string "ababab".

In the third example, you can choose the substring $s_1 s_2 s_3 s_4 s_5$ and perform the second step; then you get the string "abababa".

In the fourth example, the string is already alternating.

In the sixth example, you can choose the substring s_2 and perform the second step; then you get the string "bab".

C. Red-Black Pairs

2 seconds, 512 megabytes

There is a table of $2 \times n$ cells. Each cell of the table is colored either red or black. You want to repaint some cells of this table so that there exists at least one way to partition all cells into n pairs such that the following conditions hold:

- the cells in each pair have the same color;
- the cells in each pair share a side.

What is the minimum number of cells that need to be repainted?

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains one integer n ($1 \leq n \leq 2 \cdot 10^5$).

The second and third lines of each test case describe the colors of the cells. Each line contains a string consisting of exactly n letters "R" and/or "B".

Additional constraint on the input:

- the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output one integer — the minimum number of cells that need to be repainted.

input
5 1 R B 2 BR BR 3 RBR BRB 4 RRBB BBRB 5 RBRBR BBBRB
output
1 0 3 1 4

Consider the 3rd example. One possible option is to color all cells in one color, which requires repainting 3 cells.

In the 4th example, one possible valid repainting is

RRBB
BBRR

D. Exceptional Segments

2 seconds, 512 megabytes

You are given two integers n and x .

Consider the sequence $[1, 2, 3, \dots, n]$. You need to find the number of its subsegments that contain x and have XOR equal to 0. In other words, you need to count the number of pairs (l, r) such that $1 \leq l \leq x \leq r \leq n$ and $l \oplus (l + 1) \oplus \dots \oplus r = 0$, where $\oplus$ denotes the bitwise exclusive OR.

For example, if $n = 7$ and $x = 6$, then the following segments are suitable:

- $(4, 7)$, because x lies in this segment and $4 \oplus 5 \oplus 6 \oplus 7 = 0$;
- $(1, 7)$, because x lies in this segment and $1 \oplus 2 \oplus 3 \oplus 4 \oplus 5 \oplus 6 \oplus 7 = 0$.

Since the answer can be very large, output it modulo 998 244 353.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 2 \cdot 10^5$). The description of the test cases follows.

The only line of each test case contains two integers n and x ($1 \leq x \leq n \leq 10^{18}$).

Output

For each test case, output one integer — the number of suitable segments modulo 998 244 353.

input
5 5 5 8 1 15 8 10 10 5989566119 1996588700
output
1 2 10 0 99996

E. Covering Points with Circles

2 seconds, 512 megabytes

You are given an array p containing n points with integer coordinates. These points are uniformly distributed inside some rectangle whose sides are parallel to the coordinate axes.

You need to place several circles so that the following conditions hold:

- the radius of each circle is equal to r , and its center has integer coordinates;
- for every pair of circles, the area of their intersection is 0 (but the circles may touch);
- at least 89% of all points lie inside some circle or on the boundary of some circle (in other words, the number of points lying inside the circles or on their boundaries is at least $\frac{89n}{100}$).

The rectangle in which the points from array p are distributed is unknown to you, but in all tests except for the example it is guaranteed that the area of one circle of radius r does not exceed $\frac{1}{10}$ of the area of this rectangle.

Input

The first line contains two integers n and r ($4 \leq n \leq 10^4$, $10^2 \leq r \leq 10^3$).

The next n lines each contain two integers p_x and p_y ($-10^5 \leq p_x, p_y \leq 10^5$).

There are 40 tests in this problem. For each test except the example from the statement, the following constraints hold:

- the number of points is 10^4 ;
- all points are generated as follows: first, some integer x from 300 to 10^5 is chosen; after that, n distinct integer points are chosen uniformly at random in the rectangle $[-x, x] \times [-x, x]$;
- the area of a circle of radius r does not exceed $\frac{1}{10}$ of the area of the rectangle $[-x, x] \times [-x, x]$;
- there exists a set of circles satisfying the conditions of the problem.

Hacks are disabled in this problem.

Output

In the first line, output one integer k ($1 \leq k \leq n$) — the number of circles. In each of the next k lines, output two integers x and y — the coordinates of the center of the i -th circle. The coordinates of centers of the circles you output should not exceed $2 \cdot 10^5$ by absolute value.

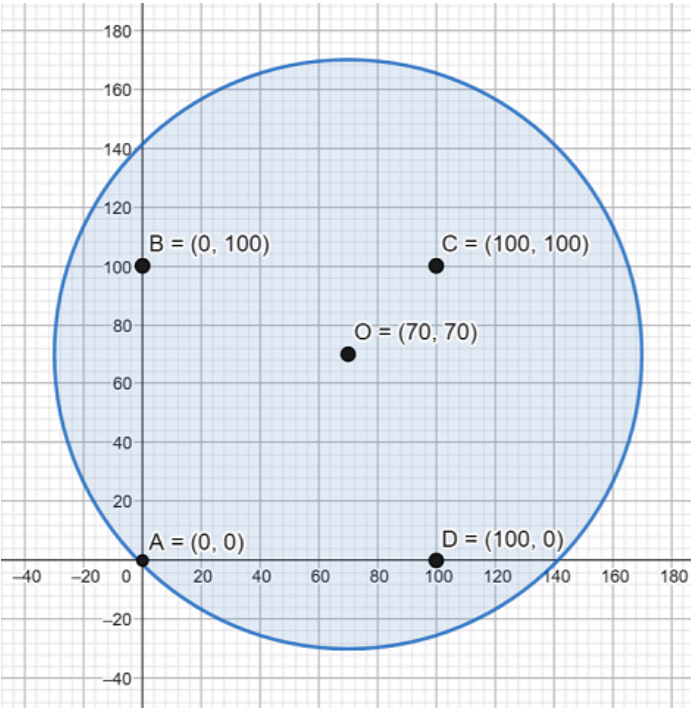
If there are multiple solutions, output any of them. You do not need to minimize the number of circles, but it should not exceed n .

input
4 100 0 0 0 100 100 0 100 100

output

1
70 70

One possible covering for the first sample is shown in the figure:



F. String Cutting

2 seconds, 512 megabytes

You are given a string s consisting of lowercase Latin letters. You have to cut it into k or more parts so that each part contains at least l letters.

After cutting the string into parts, the resulting strings will be sorted in lexicographical order. Your task is to cut the string so that the string at the k -th position in the sorted order is lexicographically as large as possible.

For example, if $s = \text{"abracadabra"}$, $l = 2$, and $k = 5$, then it can be cut into five parts:

- "ab"
- "rac"
- "ad"
- "ab"
- "ra"

After sorting the parts:

- "ab"
- "ab"
- "ad"
- "ra"
- "rac"

The string "rac" will be at the fifth position. It can be shown that this answer is maximal among all possible ways to cut the string s .

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains three integers n, l , and k ($1 \leq n, l, k \leq 10^6$).

The second line of each test case contains a string s of length n , consisting of lowercase Latin letters.

Additional constraint on the input:

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- the sum of n over all test cases does not exceed 10^6 .

Output

For each testcase, output one of the following:

- YES on the first line if an answer exists. On the second line, output the lexicographically maximum possible k -th part.
- NO on a single line if no answer exists.

Note that unlike in some other problems, **you have to print YES or NO in upper case**.

input
5 9 3 1 abcccabba 9 3 2 abcccabba 9 3 3 abcccabba 9 3 4 abcccabba 11 2 5 abracadabra
output
YES abcccabba YES ccabba YES cca NO YES rac

G. Simple Problem

2 seconds, 512 megabytes

One day, Polycarp came up with the following simple problem: the numbers $0, 1, \dots, n - 1$ are given. Is it possible to arrange them so that the absolute difference of every adjacent pair of numbers is divisible by at least one of the numbers $k_1, k_2, \dots, k_m$?

Polycarp spent a long time thinking about how to solve this problem and eventually realized that it is not so simple after all. Help him solve this problem.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^3$). The description of the test cases follows.

The first line of each test case contains two integers n and m ($3 \leq n \leq 5 \cdot 10^3; 1 \leq m \leq 10$).

The second line of each test case contains m integers k_i ($1 \leq k_i \leq \lfloor \frac{n}{3} \rfloor$).

Additional constraints on the input:

- the sum of n over all test cases does not exceed $5 \cdot 10^3$.

Output

For each test case, output one of the following:

- 1 on a single line if no suitable arrangement exists;
- n integers $0, 1, \dots, n - 1$ satisfying the condition. If there are multiple answers, print any of them.

input
2 10 2 2 3 6 1 2

output
0 2 6 8 4 1 3 5 7 9 -1

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